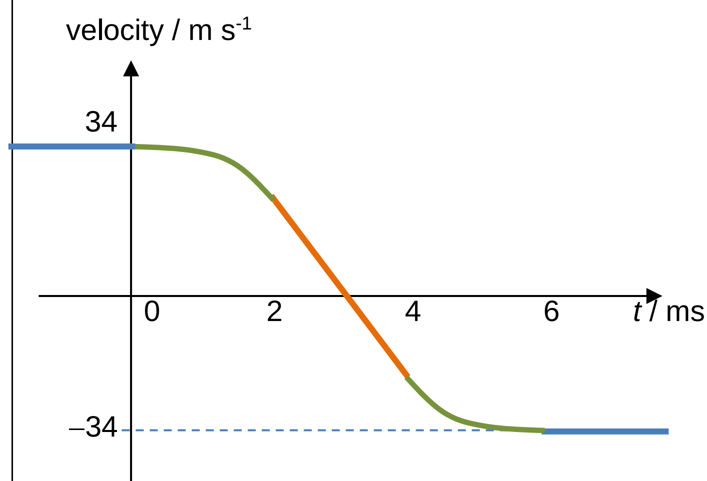
Questions			Answers	Marks
1	**(a)**		Impulse is defined as <u>product</u> of (average) force exerted on wall and time of impact	B1
	(b)		$\Delta p = 0.058 \times (-34 - 34)$	M1
			$= -3.944 \text{ kg m s}^{-1}$	A1
	(c)		impulse = change in momentum = area under $F - t$ graph	M1
			$F_{max} \times \frac{4}{1000} = 3.944$ giving $F_{max} = 986 = 990$ N (2 s.f.)	A1
	(d)		From t = 0 to t = 2 ms: decrease at increasing rate. From t = 4 ms to t = 6 ms: decrease at decreasing rate.	B1
			From t = 2 ms to t = 4 ms: decrease at constant rate.	B1
			after t = 6 ms, velocity at –34 m s^{-1}	B1
			Example	



Questions			Answers	Marks
2	(a)		molecules has component of velocity in 3 directions, hence	